

# Game Theory

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## 1 Introduction

### 1.1 Goal of Project

The goal of this project is to explore the connection between game theory and poker, specifically Texas Hold'em. We analyze how mathematical strategies, probability, and decision-making influence gameplay and how optimal strategies can be developed.

## 2 Game of Choice

### 2.1 History

Texas Hold'em is widely regarded as the most popular variant of poker, yet its exact origins remain uncertain and are still debated among historians. In 2007, the Texas State Legislature passed a resolution officially recognizing Robstown, Texas, as the birthplace of Texas Hold'em. The resolution stated that it was "traditionally held" that the first hand of the game was dealt there in the early 1900s. However, this claim relies largely on local tradition rather than documented historical evidence, as the resolution did not cite specific sources, names, or locations. As a result, while Robstown is often credited, the true origin of the game remains somewhat ambiguous.

To better understand the origins of Texas Hold'em, it is helpful to consider the broader history of card games. The standard 52-card deck has existed for several centuries, and over time, numerous poker-like games developed independently across different regions of the world. These early games spread as people traveled, particularly during long sea voyages and military campaigns, where card playing was a common form of entertainment. As a result, poker evolved gradually rather than being invented at a single moment in time.

Rather than being created by one individual, Texas Hold'em likely developed organically from earlier poker variants. Players in different regions experimented with rules, shared ideas, and adapted gameplay. This process led to many regional variations coexisting simultaneously. Because games were often played informally in saloons, homes, and social gatherings, formal documentation of rules was rare. In many cases, games were not written down until long after they had already become popular, making it difficult to trace their exact origins.

Historical accounts from professional poker players provide some insight into the early development of Texas Hold'em. Johnny Moss, one of the most famous early poker players, recalled learning the game sometime between the 1920s and 1930s in Texas, although his recollections varied over time. Similarly, legendary players Doyle Brunson and Crandell Addington both reported encountering the game in Texas during the late 1950s. Brunson referred to the game as "Hold Me Darling," and noted that its origins were rumored to be in cities such as Waco or Corpus Christi. These accounts suggest that the game was already circulating in Texas by the mid-20th century, though its exact birthplace remains unclear.

By the late 1950s and early 1960s, Texas Hold'em had spread beyond its original region and reached professional poker players. In 1963, Corky McCorquodale introduced the game to Las Vegas at the California Club. From there, it spread to a few other casinos, although it initially struggled to gain widespread popularity. For a time, the Golden Nugget was one of the only casinos where the game could be played.

Despite its limited reach, the game attracted skilled players due to its strategic depth and emphasis on decision-making.

One of the earliest widely published descriptions of Texas Hold'em appeared in a 1968 article in *LIFE Magazine*, which introduced the game to a broader American audience under names such as "Hold Me Darling" and "Texas Hold'em." The article highlighted the game's unique structure and suggested that it represented a significant development in poker. Notably, even at that time, there was little documentation of the game in earlier rule books, indicating that it was still relatively new to the mainstream.

The true breakthrough for Texas Hold'em came with the establishment of the World Series of Poker (WSOP) in 1970 by Benny and Jack Binion. This tournament provided a formal competitive setting and helped standardize the rules of the game. Over time, the WSOP grew in prestige and popularity, attracting players from around the world and solidifying Texas Hold'em as the dominant form of poker.

The game experienced another major surge in popularity during the early 2000s, often referred to as the "poker boom." This was driven by the rise of online poker platforms and televised tournaments. A key moment occurred in 2003 when amateur player Chris Moneymaker won the WSOP Main Event after qualifying through an online tournament. His victory demonstrated that ordinary players could compete at the highest level, leading to a dramatic increase in global participation.

Today, Texas Hold'em is played worldwide and is considered both a game of skill and chance. While the 2007 Texas Legislature resolution names Robstown as its birthplace, historical evidence suggests that the game evolved gradually across multiple locations in Texas. Cities such as Dallas, Fort Worth, Waco, and Corpus Christi all appear in accounts of its early development. Much like the game itself, the story of Texas Hold'em contains elements of both truth and uncertainty, making its history as complex and intriguing as the strategy it involves.

## 2.2 The Game

The objective of Texas Hold'em is to win chips either by having the best hand at showdown or by forcing all other players to fold before the hand is completed. This dual objective creates a unique balance between mathematical decision-making and psychological strategy.

Each player is dealt two private cards known as *hole cards*. These cards are visible only to the player and form the basis of their strategy. Betting proceeds clockwise around the table, beginning with the player "under the gun," who must act first with limited information.

Players have several possible actions: they may fold (give up their hand), call (match the current bet), or raise (increase the bet). These decisions depend on factors such as hand strength, position at the table, and the behavior of other players.

After the initial betting round, the dealer places three cards face-up on the table. This stage is called the *flop*, and the cards are shared by all remaining players. The introduction of community cards significantly changes the game, as players must now evaluate how their hole cards interact with the shared cards.

Following the flop, a second round of betting occurs. The dealer then reveals the fourth community card, known as the *turn*, followed by another round of betting. Finally, the fifth and last community card, the *river*, is dealt, and a final round of betting takes place.

If two or more players remain after the final betting round, a showdown occurs in which players reveal their hands. Each player forms the best possible five-card hand using any combination of their hole cards and the community cards.

One of the most important aspects of Texas Hold'em is bluffing. Because players do not always need the best hand to win, they can represent stronger hands through betting patterns, forcing opponents to fold. This makes poker fundamentally different from purely deterministic games and introduces elements of psychology and strategic deception.

## 2.3 Hand Rankings

Poker hands are ranked based on their strength, with higher-ranking hands beating lower-ranking ones. The rankings are as follows:

1. Royal Flush: The strongest possible hand, consisting of A, K, Q, J, 10 of the same suit.

2. Straight Flush: Five consecutive cards of the same suit.
3. Four of a Kind: Four cards of the same rank.
4. Full House: A combination of three of a kind and a pair.
5. Flush: Five cards of the same suit.
6. Straight: Five consecutive cards of mixed suits.
7. Three of a Kind: Three cards of the same rank.
8. Two Pair: Two distinct pairs.
9. Pair: Two cards of the same rank.
10. High Card: The highest card when no other hand is formed.

Understanding these rankings is essential, as they determine the outcome of each hand and influence strategic decisions throughout the game.

### 3 Blinds

At the beginning of each hand in poker, two players are required to place forced bets into the pot before any cards are dealt. These bets are known as the small blind and the big blind. The small blind contributes a smaller amount, while the big blind contributes a larger amount, typically equal to the minimum bet. The primary purpose of these forced bets is to ensure that there is always money in the pot, which creates an incentive for players to compete rather than fold immediately. Without this structure, players could wait indefinitely for only the strongest hands, resulting in very slow and unengaging gameplay.

The position of the blinds rotates clockwise after each hand, distributing the cost evenly among all players over time and maintaining fairness. In addition to blinds, some formats—especially tournaments—include an ante, which requires every player to contribute a small amount to the pot before the hand begins. This increases the total pot size and encourages more aggressive play, as the reward for winning becomes larger relative to the cost of participation. As a result, players must adapt by playing a wider range of hands and making more strategic decisions from the very start of each hand.

### 4 Gameplay Rounds

Poker gameplay is divided into multiple stages, each representing a different level of information available to the players. As the game progresses, more cards are revealed, allowing players to better evaluate their hands and make more informed decisions. At every stage, players must decide whether to bet, call, raise, or fold based on their hand strength, their position, and their expectations about their opponents' hands. This process involves constant adaptation, as each new card changes the set of possible outcomes and strategies.

The complexity of the game increases as more information becomes available. Early in the hand, decisions are made with limited knowledge, relying heavily on probability and general strategy. As the game advances, players can narrow down the possible hands their opponents might have and adjust their actions accordingly. By the final stage, all cards are revealed, and decisions are based on complete information about the board, though uncertainty about opponents' hidden cards still remains. This progression from uncertainty to partial clarity is what makes poker both strategically rich and mathematically interesting.

#### 4.1 Preflop

The preflop stage begins after each player receives two private cards, known as hole cards, and before any community cards are dealt. At this point, players have very limited information, as they can only see their own cards and must make decisions without knowing what their opponents hold. Because of this, preflop

decisions rely heavily on the inherent strength of the starting hand, as well as positional considerations and prior strategic knowledge.

From a mathematical perspective, preflop play involves evaluating the strength of a hand relative to all possible starting combinations. There are 1,326 distinct two-card combinations in Texas Hold'em, and each has a different likelihood of winning. Players typically group these hands into ranges that they are willing to play under certain conditions. Since uncertainty is at its highest during this stage, players must rely on probability, expected value, and strategic discipline to guide their decisions.

## 4.2 Flop, Turn, and River

The flop, turn, and river are the stages where community cards are revealed, gradually increasing the amount of information available to all players. The flop consists of three cards, the turn adds a fourth, and the river introduces the fifth and final card. With each new card, players can combine their hole cards with the shared cards to form stronger hands, while also reassessing the possible hands their opponents may have.

As more cards are revealed, the decision-making process becomes both more informed and more complex. Players must evaluate not only the current strength of their hand but also how it compares to the range of possible hands their opponents could hold. By the river, all community cards are visible, and no further information will be added. At this stage, decisions focus on maximizing value with strong hands or using bluffing strategies when appropriate, making it the most decisive and strategically nuanced phase of the game.

## 5 Game Theory Optimal (GTO)

Game Theory Optimal (GTO) poker is a strategy derived from the mathematical framework of game theory, a field that studies strategic interactions between rational decision-makers. In general, game theory analyzes situations in which the outcome for each participant depends not only on their own decisions, but also on the decisions of others. In this sense, poker is a natural application of game theory, as each player's success depends on how well they anticipate and respond to opponents.

Formally, a game in game theory consists of three main components:

- A set of players (e.g., the individuals at the poker table),
- A set of strategies available to each player (e.g., fold, call, raise, bluff),
- A payoff function that determines the outcome (profit or loss) based on all players' decisions.

Game Theory Optimal strategy is often described as the "optimal" way to play poker because it aims to maximize long-term expected profit while minimizing the possibility of being exploited by opponents. Rather than focusing on short-term outcomes, GTO emphasizes mathematically balanced play over many repeated interactions.

A central concept underlying GTO is the **Nash equilibrium**, introduced by John Nash. A Nash equilibrium is a set of strategies in which no player can improve their expected outcome by unilaterally changing their strategy while others keep theirs fixed. In the context of poker, this means that if all players are using GTO strategies, no player can gain a consistent advantage over the others.

In GTO poker, players construct **balanced ranges** of hands rather than playing individual hands in isolation. Each possible starting hand (such as AA, KK, or AK) has a certain probability of winning, and these probabilities are used to determine how frequently each hand should be played aggressively, passively, or folded. Importantly, strong hands are not the only hands played aggressively; weaker hands are also included in betting ranges at certain frequencies to maintain unpredictability. This concept is closely related to **mixed strategies**, where players randomize their decisions to prevent opponents from identifying patterns.

For example, a player may choose to bluff with a weak hand a certain percentage of the time, even when it is not immediately profitable. This ensures that opponents cannot simply assume that every bet represents a strong hand. Mathematically, this creates a balance where opponents are indifferent between calling and folding, which is a hallmark of equilibrium play.

Another key pillar of game theory present in poker is the concept of **best response**. A best response is the strategy that yields the highest payoff given the strategies chosen by other players. In GTO play, strategies are constructed so that opponents cannot find a better response that consistently exploits them.

Poker can also be modeled in two different ways within game theory:

- **Normal form games**, where all strategies and payoffs are represented in a matrix,
- **Extensive form games**, where decisions occur sequentially and are represented as a game tree.

Texas Hold'em is best described as an extensive form game because players make decisions over multiple stages (preflop, flop, turn, river), each with increasing information. At each stage, players must update their strategies based on new information, making the game highly dynamic.

In real-world poker, however, players do not always follow optimal strategies. This introduces the idea of **exploitative play**, where a player deliberately deviates from GTO in order to take advantage of opponents' mistakes. For example, if an opponent folds too often, a player can profit by bluffing more frequently than GTO would suggest. Conversely, if an opponent calls too often, the player can reduce bluffs and focus on value betting strong hands.

This creates a constantly evolving strategic environment. As one player adjusts their strategy to exploit another, the opponent may respond by adjusting in return. This cycle reflects a fundamental principle of game theory: strategic interdependence.

In heads-up poker (a two-player game), researchers have been able to approximate GTO strategies with high precision. In 2015, a team from the University of Alberta used over 200 computers running for several weeks to effectively "solve" heads-up limit Texas Hold'em. Their algorithm learned by playing millions of simulated hands against itself, gradually converging to a strategy that is nearly impossible to exploit.

One significant result of this research was the quantification of positional advantage. The player in the dealer position (acting last) was found to have an expected advantage of approximately 0.088 big blinds per hand. While this may seem small, over thousands of hands it represents a substantial edge, illustrating how even minor probabilistic advantages accumulate over time.

More broadly, poker serves as a powerful real-world example of game theory in action. Each decision involves uncertainty, incomplete information, and strategic interaction with other players. This is similar to many economic scenarios, such as auctions, where participants must make decisions based on both their own valuation and expectations of others' behavior.

In conclusion, GTO poker represents the intersection of mathematics, probability, and strategic reasoning. It demonstrates how abstract concepts such as Nash equilibrium, mixed strategies, and payoff optimization can be applied to real-world decision-making. While perfect GTO play is rarely achievable in practice, it provides a theoretical foundation upon which modern poker strategy is built.

## 6 Poker Ranges

A poker starting hand range chart is a visual representation of all possible two-card combinations in Texas Hold'em and is an essential tool in GTO strategy.

**Card matrix**

|     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| AA  | AKs | AQs | AJs | ATs | A9s | A8s | A7s | A6s | A5s | A4s | A3s | A2s |
| AKo | KK  | KQs | KJs | KTs | K9s | K8s | K7s | K6s | K5s | K4s | K3s | K2s |
| AQo | KQo | QQ  | QJs | QTs | Q9s | Q8s | Q7s | Q6s | Q5s | Q4s | Q3s | Q2s |
| AJo | KJo | QJo | JJ  | JTs | J9s | J8s | J7s | J6s | J5s | J4s | J3s | J2s |
| ATo | KTo | QTo | JTo | TT  | T9s | T8s | T7s | T6s | T5s | T4s | T3s | T2s |
| A9o | K9o | Q9o | J9o | T9o | 99  | 98s | 97s | 96s | 95s | 94s | 93s | 92s |
| A8o | K8o | Q8o | J8o | T8o | 98o | 88  | 87s | 86s | 85s | 84s | 83s | 82s |
| A7o | K7o | Q7o | J7o | T7o | 97o | 87o | 77  | 76s | 75s | 74s | 73s | 72s |
| A6o | K6o | Q6o | J6o | T6o | 96o | 86o | 76o | 66  | 65s | 64s | 63s | 62s |
| A5o | K5o | Q5o | J5o | T5o | 95o | 85o | 75o | 65o | 55  | 54s | 53s | 52s |
| A4o | K4o | Q4o | J4o | T4o | 94o | 84o | 74o | 64o | 54o | 44  | 43s | 42s |
| A3o | K3o | Q3o | J3o | T3o | 93o | 83o | 73o | 63o | 53o | 43o | 33  | 32s |
| A2o | K2o | Q2o | J2o | T2o | 92o | 82o | 72o | 62o | 52o | 42o | 32o | 22  |

Figure 1: Poker starting hand range chart used in GTO strategy

Each square in the chart represents a specific hand. Diagonal entries correspond to pocket pairs, while hands above the diagonal are suited and those below are offsuit.

The chart uses colors to indicate how each hand should be played. Strong hands are typically raised, medium-strength hands may be called, and weaker hands are folded. This creates a balanced strategy that prevents opponents from easily predicting a player's actions.

This chart illustrates a fundamental concept in game theory: balance. If a player only plays strong hands, opponents can exploit this by folding whenever the player shows aggression. By mixing strategies and including weaker hands, players become more unpredictable and harder to exploit.

Although range charts provide a mathematically optimal baseline, real gameplay often requires adjustments based on opponents' tendencies. As a result, poker combines elements of mathematics, probability, and psychology, making it a rich subject for analysis within game theory.

## 7 Poker Math

### 7.1 Binomial Coefficient

The binomial coefficient  $\binom{n}{k}$  counts the number of ways to choose  $k$  elements from a set of  $n$  distinct elements. This is done when order does not matter and no element is repeated.

## Binomial Coefficient

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}, \text{ given } n \geq 0 \text{ and } k \geq 0$$

**$n$ :** Total number of elements

**$k$ :** Number of elements in the desired subset

### **Derivation**

The goal is to find the number of ways  $k$  elements can be chosen from a set of  $n$  distinct elements. This number is written as  $\binom{n}{k}$ .

1. Count permutations of all  $n$  elements

Arranging all  $n$  distinct elements in order gives:

$$n! = n(n-1)(n-2) \cdots 1$$

This counts every possible ordering of the entire set.

2. Overcounting from ordering the chosen elements

If we fix a particular subset of  $k$  chosen elements, those  $k$  elements can be arranged among themselves in:

$$k! = k(k-1)(k-2) \cdots 1$$

All of these orderings represent the same subset, since combinations ignore order.

3. Overcounting from ordering the non-chosen elements

The remaining  $n-k$  elements can also be arranged among themselves in:

$$(n-k)! = (n-k)(n-k-1)(n-k-2) \cdots 1$$

These orderings also do not change which subset of size  $k$  was chosen.

4. Total overcounting per subset

For each subset of size  $k$ , every ordering of the chosen elements can be paired with every ordering of the non-chosen elements. Therefore, each subset corresponds to:

$$k!(n-k)!$$

different full permutations of the  $n$  elements.

5. Final formula

To obtain the number of distinct subsets of size  $k$ , divide the total number of permutations by the number of permutations corresponding to each subset:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

In expanded form:

$$\binom{n}{k} = \frac{n(n-1)(n-2) \cdots (n-k+1)}{k(k-1)(k-2) \cdots 1}$$

(Rosen, 2019, Section 6.5)

## 7.2 Poker Hand Probabilities

The hierarchy of each poker hand is ranked by the increasing improbability of each hand occurring.

### Total 5-Card Combinations

- standard 52 card deck
- 5 card hands
- No card repeats

$$\binom{52}{5} = \frac{52!}{5!(52-5)!} = 2,598,960$$

$n$ : Total number of cards in a deck

$r$ : 5 card hand

#### 1. Royal Flush

A Royal Flush requires the sequence A,K,Q,J,10 all sharing the same suit. This leaves only four possible combinations, one for each suit.

##### Possible Hands

$$\binom{4}{1} = 4$$

##### Hand Probability

$$\frac{4}{2,598,960} = 0.00015\%$$

#### 2. Straight Flush

While only one is required, there exists 10 possible sequential combinations of five cards holding the same suit. However, one of those is considered the Royal flush, leaving only nine. This is multiplied with the possibility of four possible suits but only needing one.

##### Possible Hands

$$\binom{9}{1} \binom{4}{1} = 9 * 4 = 36$$

##### Hand Probability

$$\frac{36}{2,598,960} = 0.0014\%$$

#### 3. Four of a Kind

Since four of a kind only requires four cards, the number of possible hands is elevated to 10. As now cards from a Royal Flush can be included. Once again, this is spread across the four suits.

##### Possible Hands



$$\binom{13}{1} \binom{4}{4} \binom{12}{1} \binom{4}{1} = 624$$

#### Hand Probability

$$\frac{40}{2,598,960} = 0.024\%$$

#### 4. Full House

This hand consists of one Three-of-a-kind and a one pair combination, where each card in each combination are of the same rank. Given there are four cards to a rank, each card combination must belong to a different rank. This leaves 13 possible ranks for the first combination and 12 for the second.

##### Possible Hands

$$\binom{13}{1} \binom{4}{3} \binom{12}{1} \binom{4}{2} = 3,744$$

#### Hand Probability

$$\frac{3744}{2,598,960} = 0.1441\%$$

#### 5. Flush

A Flush requires five cards the of same suit. However, this hand excludes the card combinations of a Royal and Straight Flush as these hands are entirely different. With 1 combination per suit for a Royal Flush, and nine for a Straight Flush, a total of 10 hands per suit must be excluded.

##### Possible Hands

$$\binom{13}{5} \binom{4}{1} - \binom{10}{1} \binom{4}{1} = 5,108$$

#### Hand Probability

$$\frac{5,108}{2,598,960} = 0.197\%$$

#### 6. Straight

The Straight is a hand in which all five cards are in sequential order regardless of suit. However, a lesser hand cannot account for one with greater standing. So, the Royal Flush and Straight Flush must be excluded. As they are the only other hands with five cards in sequential order. There are 10 possible Straights, and each  $\binom{4}{1}$  accounts for a single card. With four representing the suits and one representing the amount of cards from that rank required for the hand.

##### Possible Hands

$$\binom{10}{1} \binom{4}{1} \binom{4}{1} \binom{4}{1} \binom{4}{1} \binom{4}{1} - \binom{10}{1} \binom{4}{1} = 10,200$$

#### Hand Probability

$$\frac{10,200}{2,598,960} = 0.393\%$$

#### 7. Three of a Kind

Just as its name indicates, this hand simply requires three equivalently ranked cards to be selected. This allows the remaining two cards to assume any rank, as long as it differs from the first three, and any suit.

##### Possible Hands

$$\binom{13}{1} \binom{4}{3} \binom{12}{2} \binom{4}{1} \binom{4}{1} = 54,912$$

### Hand Probability

$$\frac{54,912}{2,598,960} = 2.11\%$$

#### 8. Two Pair

To satisfy Two Pair, two cards must be the same, two separate times. For example, two sixes and two 10s, with each pair assuming a different rank. Otherwise, the hand would be four-of-a-kind. The unpaired card can be any suit and any rank not assumed by a pair.

##### Possible Hands

$$\binom{13}{1} \binom{4}{2} \binom{12}{1} \binom{4}{2} \binom{11}{1} \binom{4}{1} = 123,552$$

### Hand Probability

$$\frac{123,552}{2,598,960} = 4.75\%$$

#### 9. Pair

Just like above, only one pair now.

##### Possible Hands

$$\binom{13}{1} \binom{4}{2} \binom{12}{1} \binom{4}{1} \binom{11}{1} \binom{4}{1} \binom{10}{1} \binom{4}{1} = 1,098,240$$

### Hand Probability

$$\frac{1,098,240}{2,598,960} = 42.26\%$$

#### 10. High Card

When a hand has absolutely no combinational value, only high card remains. This requires a player to place the value of their hand in the hopes of their highest ranked card.  $\binom{13}{5}$  is the selection of 5 cards all of varying rank, eliminating the possibility of a pair. However, Straights and Flushes still remain possible.  $\binom{13}{5} - \binom{10}{1}$  removes the possibility of a Straight, so it is the measure of picking cards of non-consecutive rank.  $\binom{4}{1}^5 - \binom{4}{1}$  represents the removal of a flush, preventing cards from all sharing the same suit.  $\binom{4}{1}$  represents how for every rank to get a flush one suit must be chosen. But by removing the selection of a single suit, it counteracts the possibility of all cards sharing a suit.

##### Possible Hands

$$(\binom{13}{5} - \binom{10}{1})(\binom{4}{1}^5 - \binom{4}{1}) = 1,302,540$$

### Hand Probability

$$\frac{1,302,540}{2,598,960} = 50.12\%$$

(Poker Probability, 2026)

## 7.3 Outs

With countless unshared information between players, its important to calculate the probability of unknown future events to have better winning odds. The main technical approach used by players is called counting outs. Counting outs emphasizes offensive play, focusing on avenues to improve the success of your current hand and thus win more money (Feterik, 2026). In Texas Hold'em, after the initial deal of two private cards (Hole cards) to each player, three cards are laid face up on the table for all to see. These first three public cards are called the flop. After the flop is dealt, counting outs begins (Feterik, 2026). Counting outs involves analyzing what the *next* card dealt needs to be so that your current hand improves (Feterik, 2026). It is important to note counting outs does not include evaluating the combinational probabilities of better hands occurring. So, requiring *two* cards to craft a better hand does *not* count as an out.

| Symbol | Meaning  |
|--------|----------|
| C      | Clubs    |
| S      | Spades   |
| H      | Hearts   |
| D      | Diamonds |

Card Suit Key

Turn 1

| Hand Name | Card     |
|-----------|----------|
| Hole      | QH 5S    |
| Flop      | KS 5C 6H |

**Example:** Simulated Poker Hand

### Calculating Turn 1 Outs:

Two pair: After the flop, a pair of fives is present. With the five of Spades coming from the Hole and the five of Clubs from the flop. Given this, the first out to consider is the two pair. As it is the first hand to outplay the pair and the most plausible improvement. Given there are three other cards present, these are the three to focus on.

Outs  
 $3 \binom{3}{1}$  or  $\binom{9}{1}$ : Three cards in the deck remain for each King, Queen, and Six. But only one card is required to make another pair.

$$\binom{9}{1} = 9 \text{ outs}$$

Three of a kind: After considering a two pair, the next hand is three of a kind. Given only one pair currently exists, three fives is the only possible pursuit that can be achieved in the next turn.

Outs  
 $\binom{2}{1}$ : Only two fives remain to be drawn.

$$\binom{2}{1} = 2 \text{ outs}$$

Turn 2

| Hand Name | Card |
|-----------|------|
| Turn      | 2S   |

### Calculating Turn 2 Outs

A two is not desired for any of the hands. This eliminates the possibility of:

Pair: Given a two has no practical application to any of the outs, a pair remains as the current playable hand.

Two Pair: A Two Pair remains possible, as the River only needs to yield a single King, Queen, Six, or Two to create a second pair

Outs  
 $\binom{12}{1}$ : Each King, Queen, Six, and Two has three remaining cards in the deck. Only one is needed to complete another pair.

$$\binom{12}{1} = 12 \text{ outs}$$

Three of a Kind: While drawing a two certainly decreases the probability of three of a kind it is still possible. The River must yield a five in the final draw to achieve this hand.

Outs

$\binom{2}{1}$ : One of the two remaining 5s needs to be drawn to complete the Three-of-a-kind.

$$\binom{2}{1} = 2 \text{ outs}$$

Turn 3

| Hand Name | Card |
|-----------|------|
| River     | KH   |

### Turn 3

Two Pair: Given a King has been drawn, the only possible hand is a two pair. However, because a two pair is better than a single pair, this round can be seen as a relative success.

## 7.4 Poker Ranges

Suited: Both cards share the same suit

Off-Suit: Each card has a different suit

### Total Possible Hole Card Strength Combinations:

13 ranks, 2 cards selected on two separate trials.

$$\binom{13}{1}^2 = 169$$

**Note**: Due to the varying perspectives on Strong, Moderate, and Weak hands, the following probabilities were derived simply through counting. If you google Poker Ranges, you will find that there are varying levels of complexity in the maps. Which would directly affect the probability of each strength of hand. For generalizing purposes, our poker range only has three rankings.

Probability of a Strong Hand

$$\frac{40}{169} = 23.67\%$$

Probability of a Mid-Level Hand

$$\frac{56}{169} = 33.14\%$$

Probability of a Poor Hand

$$\frac{73}{169} = 43.195\%$$

**Hole Card Hands When Considering Suit:**

**Total Possible Two Card Hands:**

$$\binom{52}{2} = 1,326$$

Pocket Pair Probability:

$\binom{13}{1}$ : One rank chosen for both cards.

$\binom{4}{2}$ : Two suits chosen, one for each card.

$$\frac{\binom{13}{1}\binom{4}{2}}{\binom{52}{2}} = 5.88\%$$

Possible Variations per Pair:

Given you are dealt two cards of the same rank, there are 78 possible hands you could be playing with.

$$\binom{13}{1}\binom{4}{2} = 78$$

Probability of an Off-Suit Hand:

$\binom{13}{2}$ : Pairs are excluded by picking cards of two different ranks.

$\binom{4}{1}^2$ : Each of the two cards have four possible suit pairings.

$-\binom{4}{1}$ : Represents the removal of the four possibilities in which two cards share the same suit.

$$\frac{\binom{13}{2}(\binom{4}{1}^2 - \binom{4}{1})}{\binom{52}{2}} = 70.59\%$$

Possible Variations for Every Off-Suited Hand:

Given you are dealt two cards with differing suits, there are 936 possible hands you could be playing with.

$$\binom{13}{2}(\binom{4}{1}^2 - \binom{4}{1}) = 936$$

Probability of an On-Suit Hand:

$\binom{13}{2}$ : To share a suit, two cards must be of separate rank.

$\binom{4}{1}$ : Of four possible suits, chose the same suit for both.

$$\frac{\binom{13}{2}\binom{4}{1}}{\binom{52}{2}} = 23.53\%$$

Possible On-Suit Variations:

Given you are dealt two cards of the same suit, there are 312 possible hands you could be playing with.

$$\binom{13}{2}\binom{4}{1} = 312$$

## 7.5 Betting

When placing a bet, once money enters the pot the possibility of earning a portion of the total is ended. As the round can only conclude with an absolute winner and loser. However, over time a player must maximize the profitability of their play by analyzing profit and loss in every situation (Bos, 2022). This can be done by evaluating both the expected theoretical proportion of the pot a player could win and the monetary value of a bet at a given time.

### **Expected Value**

*EV*: Expected dollar value won or lost in the long run.

*W%*: Probability of winning.

*L%*: Probability of losing.

*\$W*: Possible earnings given a win.

*\$L*: Possible losses given a loss.

$$EV = (W\% * \$W) - (L\% * \$L)$$

Interpretation: Once the EV has been calculated, a positive EV signifies a player should bet. Essentially stating that over countless trials this specific hand given this amount of potential earnings results in long

term profitability (Bos, 2022). While a negative EV says the opposite.

### Finding \$W and \$L

Calculating the potential amount won and lost for a specific bet is quite simple. Amount won (\$W) is the size of the pot should the bet be made. While amount lost (\$L) is the bet amount (Bos, 2022).

**Example** Assume the current pot is worth \$10. The player to your right raises \$5 to increase the pot to \$15. This is when you decide you want to calculate the Expected Value of raising to determine its profitability. Given a \$5 call, this puts \$W at 20, as \$W is the sum of the current pot (\$15) and your potential call of \$5. \$L meanwhile is just \$5 as it represents the wager amount that could be lost.

### Poker Equity

Poker equity is the proportion of the pot that should belong to one player. An evaluation of what *should* be earned but not a measure of what *will* be earned (Poker.org, 2024). While masked in financial terms, equity is simply the probability a hand will win. In contrast, Expected Value is the long term monetary value of a decision.

The most common strategy to calculate win and loss probabilities is called the 4-2 rule. This rule states that after the flop has been dealt but before the turn has been dealt the rule of 4 can be applied. This means that if the number of outs for a given hand is multiplied by four the rough winning percentage can be found for that hand. Once the turn has been dealt, the number outs for a given hand can be multiplied by two to reveal the winning percentage (Poker.org, 2024). The 4-2 rule originates from the idea that after the flop but before the turn only five cards have been dealt. This leaves two cards undealt, with the probability of the next two cards being  $\frac{1}{47} + \frac{1}{46}$ , which rounds to about 4% (BetMGM, 2023). The multiplication of the 2 follows the same idea, between the turn the river, one card still needs to be dealt. This leaves the probability of the draw at  $\frac{1}{46}$  which is roughly 2% (BetMGM, 2023). This is then multiplied with the outs to find the probability of the next card or one of the next two being a card that can improve the current hand.

However, after the river is dealt the 4-2 rule expires. This requires win probability to be found by looking at the number of five card permutations that are of lesser value than your current hand. Once the winning percentage is found the loss percentage can easily be calculated with  $1 - W\%$ .

#### Pre-Turn Deal

X: Number of outs for a given hand.

$$W\% = 4X$$

$$L\% = 1 - W\%$$

#### Post-Turn Deal

X: Number of outs for a given hand.

$$W\% = 2X$$

### Example

For this example we will use the simulated outs of the Two Pair from above.

#### Turn 1:

After the flop, to get another pair one more King, Queen, or Six needs to be drawn. This leaves 9 possible outs. As three of each card remain undrawn in the deck.

$$W\% = 4 * 9 = 36\%$$

$$L\% = 1 - 0.36 = 64\%$$

#### Turn 2:

The turn reveals a two, raising the total outs to 12. As now three Kings, Queens, Sixes, and Twos remain in the deck waiting to be drawn to complete the two pair.

$$W\% = 2 * 12 = 24\%$$

$$L\% = 1 - 0.24 = 76\%$$

### Turn 3:

The river revealed a King, allowing for the final hand to be a two pair. A two pair of King high is a relatively strong hand, but to be for sure we need to find the winning percentage. First, it is important to note that when evaluating the strength of a pair the highest pair wins. So, a King-Five two pair would outweigh a Jack-Queen two pair as the King out ranks the Queen.

Total permutations of lesser value:

All inferior Two pairs:

$\binom{11}{2}\binom{4}{2}^2$ : After Ace and King are excluded, there are 11 remaining inferior ranks for four cards to assume.

Pair:

$\binom{13}{1}\binom{4}{2}\binom{12}{3}\binom{4}{1}^3$

High Card:

$\binom{13}{5} - \binom{10}{1}(\binom{4}{1}^5 - \binom{4}{1})$

$$\binom{11}{2}\binom{4}{2}^2 + \binom{13}{1}\binom{4}{2}\binom{12}{3}\binom{4}{1}^3 + (\binom{13}{5} - \binom{10}{1})(\binom{4}{1}^5 - \binom{4}{1}) = 2,401,110$$

$$W\% = \frac{2,401,110}{2,598,960} = 92.38\%$$

$$L\% = 1 - 0.9238 = 7.61\%$$

### Expected Value:

Combining the two examples previously mentioned, lets evaluate the profitability of the \$5 call in a \$20 pot while possibly holding a two pair in turn 1.

$$EV = (20 * .36) - (5 * .64) = 4$$

A positive EV declares the \$5 to be a financially responsible bet. Now lets move to round 2, where the pot is now \$150 and you're being asked to call \$12.

$$EV = (162 * 0.24) - (12 * 0.76) = 49.2$$

Compared to the previous EV of 4, this is an extremely strong case pushing you to bet. Now that the EV has pushed bets in both previous rounds, we will be playing in a third betting round. The river revealed a second two, giving us a two pair of King-Five. Lets imagine the pot is now at \$225 with a call of \$20 being demanded.

$$EV = (245 * 0.9238) - (20 * 0.0761) = 224.809$$

Unsurprisingly the EV is astoundingly positive, signifying this is a good bet. This rationale will continue until either the EV says otherwise or the bets will close ending the round. Thus forcing people to reveal their cards to decide the round winner.

## 8 Relevant Game Theory

Game theory is the mathematical study of strategic interactions between rational decision-makers, where the outcome for each participant depends not only on their own actions, but also on the actions of others. In the context of poker, this framework is particularly useful because players must make decisions under uncertainty while anticipating the behavior of their opponents.

Texas Hold'em can be modeled as a game of incomplete information, since players do not have access to their opponents' private cards. Instead, they form beliefs about possible hands and act accordingly. This type of setting is commonly modeled as a Bayesian game, where players make optimal decisions based on probabilities and expectations.

A central concept in game theory is the expected value (EV), which represents the average payoff of a strategy weighted by the probabilities of different outcomes. Rational players aim to maximize expected value over the long run rather than focusing on individual outcomes.

Another key concept is that of mixed strategies, in which players randomize their actions with certain probabilities. In poker, this corresponds to actions such as bluffing with a weak hand a specific percentage of the time in order to remain unpredictable. Without randomization, opponents could exploit predictable behavior.

The primary equilibrium concept used in this analysis is the Nash equilibrium. A Nash equilibrium is a set of strategies such that no player can improve their expected payoff by unilaterally deviating from their strategy. In poker, this corresponds to balanced play where bluffing and calling frequencies are chosen so that opponents are indifferent between their available actions.

## 8.1 Game Representation

We analyze a simplified two-player model of Texas Hold'em in which Player 1 (the bettor) can either hold a strong hand (S) or bluff (B), and Player 2 can either call (C) or fold (F). The payoffs are determined by a simplified betting scenario with a fixed pot and bet size.

This interaction can be represented in normal form by the following payoff matrix:

|          | <i>C</i>    | <i>F</i>    |
|----------|-------------|-------------|
| <i>S</i> | (150, -150) | (100, -100) |
| <i>B</i> | (-50, 50)   | (100, -100) |

Here, the first entry in each pair represents Player 1's payoff, while the second represents Player 2's payoff. These values are based on a simplified model in which winning the pot yields a positive payoff and losing a call results in a negative payoff. Folding results in a smaller, fixed loss.

## 8.2 Derivation of Equilibrium

To determine the optimal strategies, we solve for a mixed-strategy Nash equilibrium. Let:

- $p$  = probability that Player 1 bluffs,
- $q$  = probability that Player 2 calls.

### 8.2.1 Player 2's Decision

If Player 2 calls, their expected value is:

$$EV_2(C) = p(50) + (1 - p)(-150)$$

$$EV_2(C) = 200p - 150$$

If Player 2 folds, their expected value is:

$$EV_2(F) = -100$$

Setting  $EV_2(C) = EV_2(F)$  to find indifference:

$$200p - 150 = -100$$

$$200p = 50$$

$$p = 0.25$$



### 8.2.2 Player 1's Decision

If Player 1 bluffs, their expected value is:

$$EV_1(B) = q(-50) + (1 - q)(100)$$

$$EV_1(B) = 100 - 150q$$

If Player 1 has a strong hand, their expected value is:

$$EV_1(S) = q(150) + (1 - q)(100)$$

$$EV_1(S) = 100 + 50q$$

Setting  $EV_1(B) = EV_1(S)$ :

$$100 - 150q = 100 + 50q$$

$$-150q = 50q$$

$$q = 0.25$$

### 8.3 Nash Equilibrium

The mixed-strategy Nash equilibrium occurs when: [  $p = 0.25$ ,  $q = 0.25$  ]

At this point, both players are indifferent between their available strategies, and neither player can improve their expected payoff by changing their strategy alone. This results in a balanced strategy in which Player 1 bluffs 25

### 8.4 Interpretation in Poker

This result illustrates a fundamental principle of poker strategy: optimal play requires balance. If Player 1 bluffs too frequently, Player 2 can exploit this by calling more often. Conversely, if Player 1 never bluffs, Player 2 can safely fold against bets. The equilibrium ensures that both players remain unpredictable and prevents either from gaining a systematic advantage.

Although this model is highly simplified, it captures the essential strategic structure of poker and demonstrates how game theory can be used to determine optimal decision-making under uncertainty.

## 9 Future Areas of Study

Game Theory is interesting because it can be applied to such a wide variety of situations. In the beginning of this project we considered studying the game theory of a game called the Oregon Trail. The Oregon Trail is a role play game in which a player tries to travel from the east coast to the west coast on the Oregon Trail. In their journey a player will encounter a variety of hazards like illness, broken wheels on their wagon, river crossings, and so on. Every decision a player makes directly impacts their future success. For example, at a river crossing a player may choose to wade across instead of looking for safer passage upstream. This could cause the wagon to tip and lose valuable supplies which could increase the chance of future sicknesses. For every decision there is a possible repercussion and trade off. While it is a lighthearted game, it would be interesting to see if there is any optimal strategy for navigating the Oregon Trail and its many hazards.

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